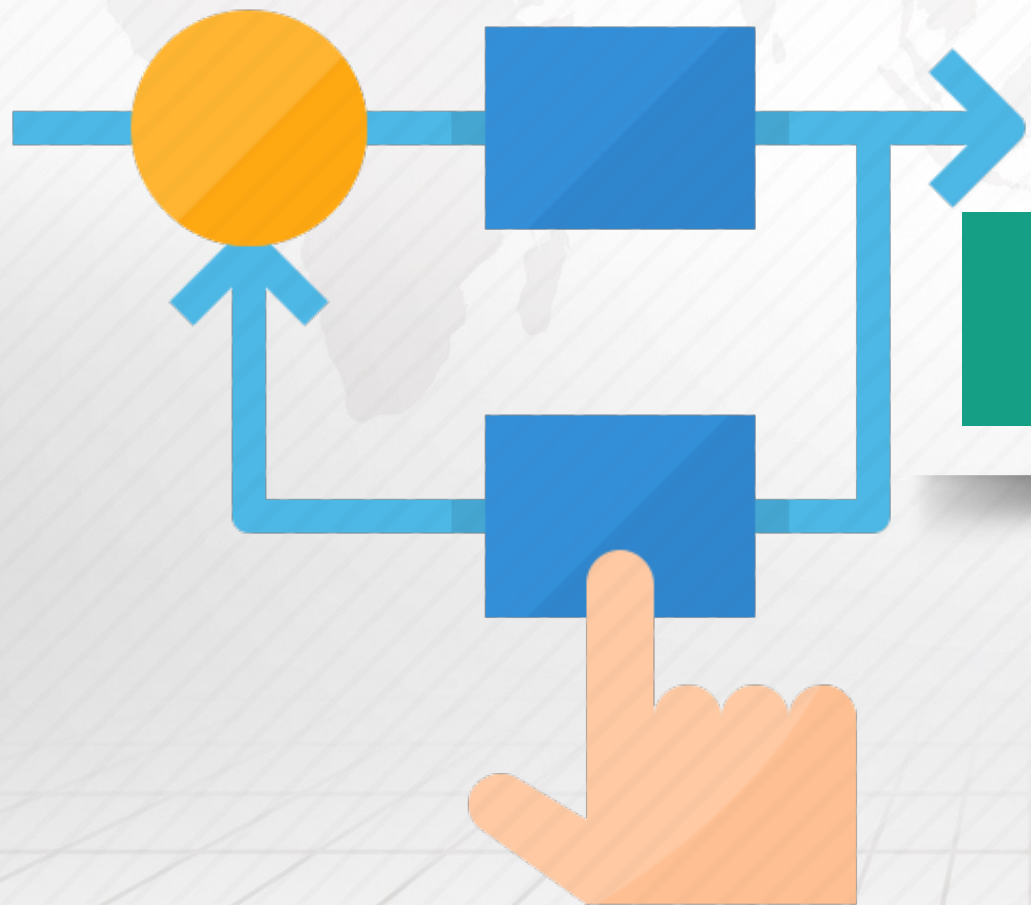




江苏科技大学
JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY



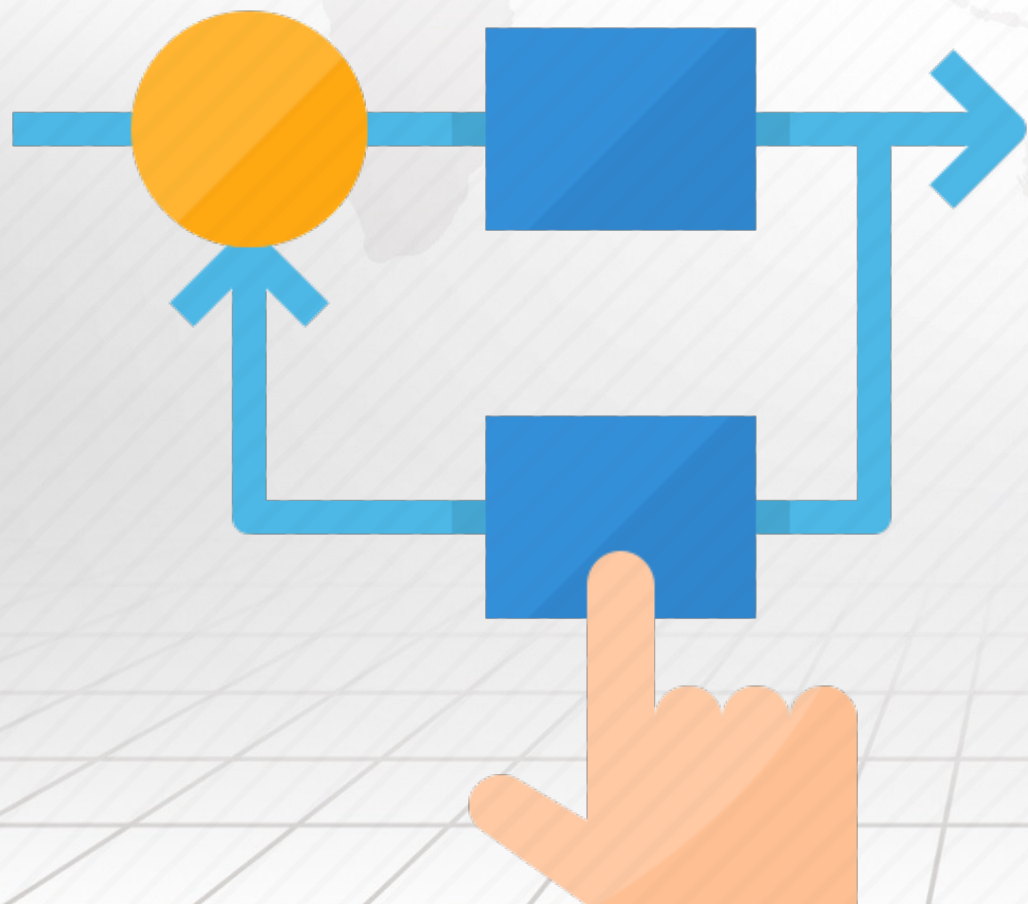
自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

22. 超前校正：幅加角增



学习目标



01

能解释超前网络典型特性

重

能运用超前校正设计控制器

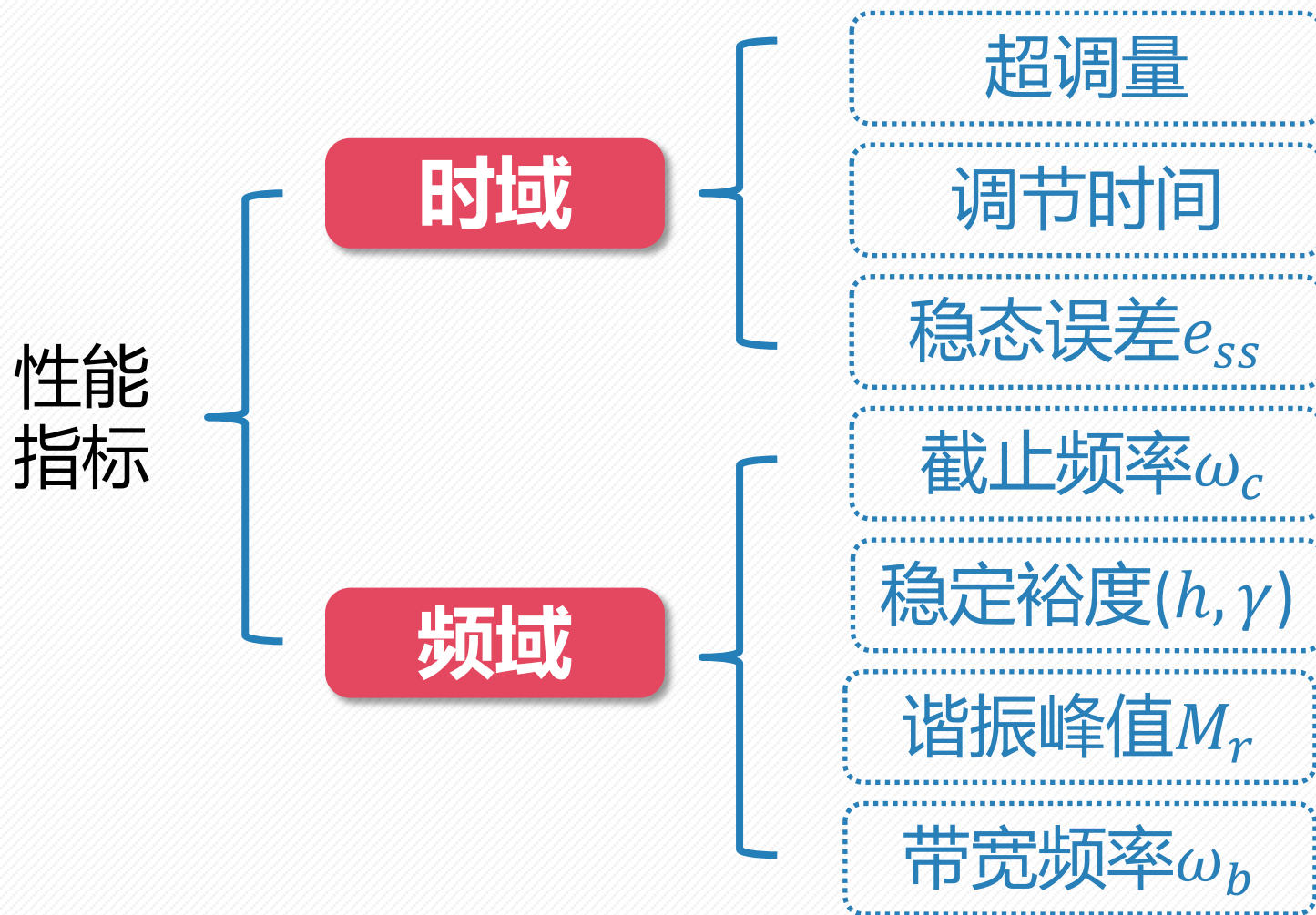
难

能综合运用计算机辅助分析系统

04

能通过估算简化设计过程

控制目标-性能指标



控制目标-性能指标

开环频率特性



系统性能指标

1 暂态响应好，稳态误差大



增加低频增益减小稳态误差，保持中、高频特性不变

2 稳态误差小，动态性能差



改变中高频以改变截止频率和相角裕度

3 稳态暂态均不满意

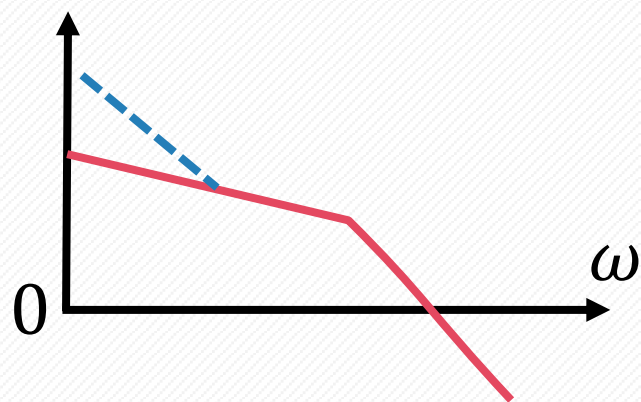
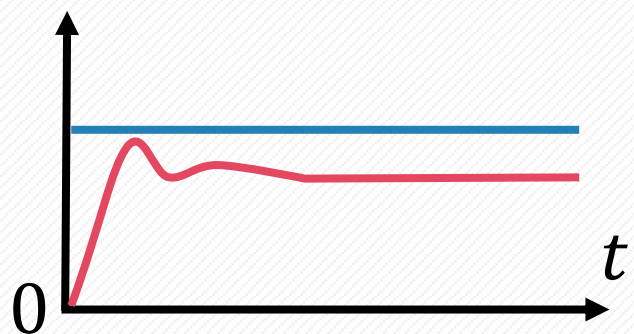


增加低频增益，改变中高频段

控制目标-性能指标

1

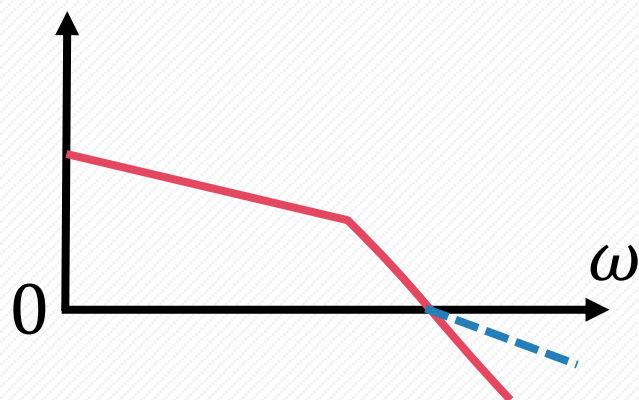
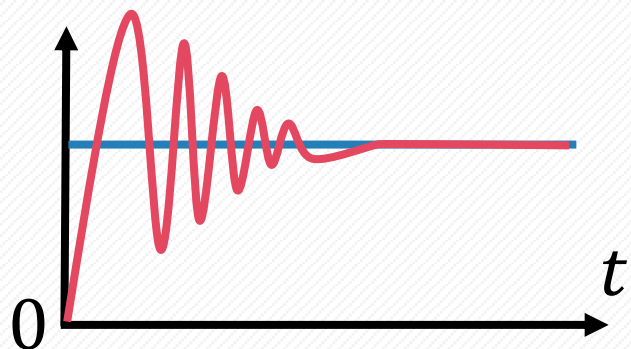
暂态响应好
稳态误差大



改变低频段

2

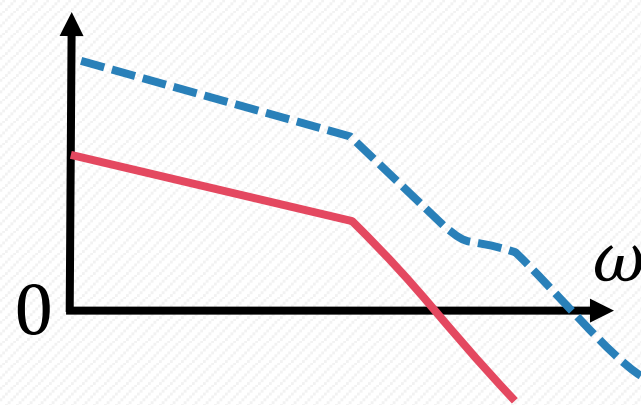
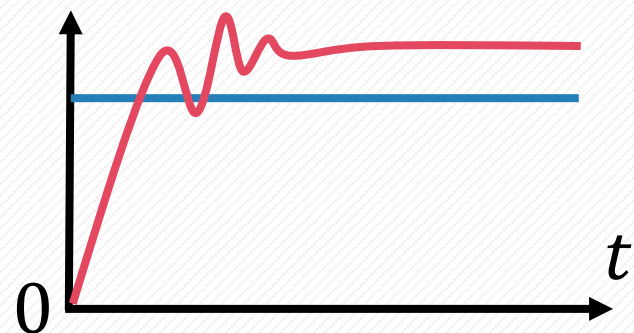
稳态误差小
动态性能差



改变中高频段

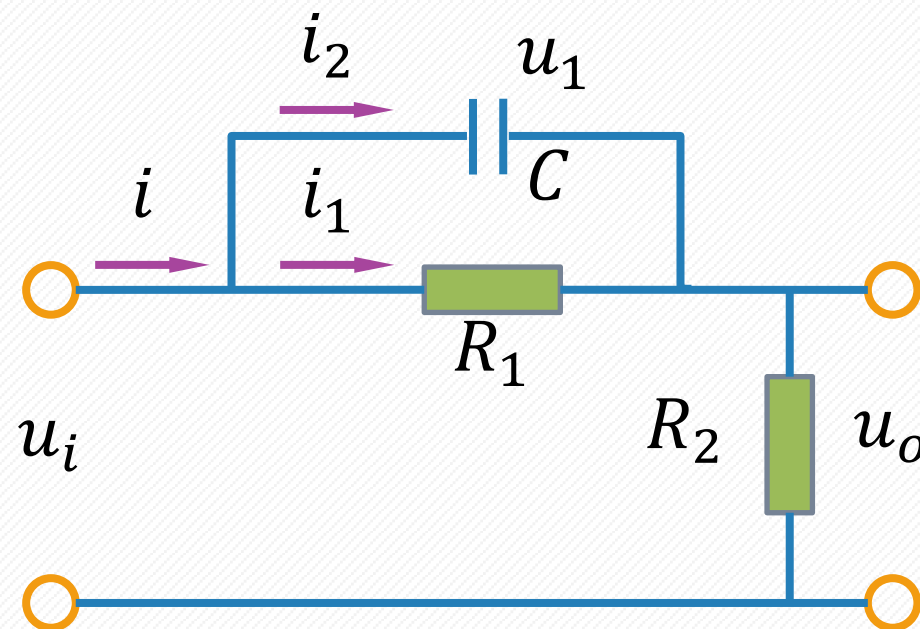
3

稳态暂态
均不满意



低中高频段均改变

超前网络特性



$$G_s(s) = \frac{U_o}{U_i} = \frac{CR_2s + \frac{R_2}{R_1}}{CR_2s + \frac{R_2}{R_1} + 1} = \frac{1}{a} \cdot \frac{1 + aTs}{1 + Ts}$$

$$T = \frac{R_1 R_2 C}{R_1 + R_2}$$

$$a = \frac{R_1 + R_2}{R_2}$$

$$a \cdot G_s(s) = \frac{aTs + 1}{Ts + 1}$$

超前网络特性

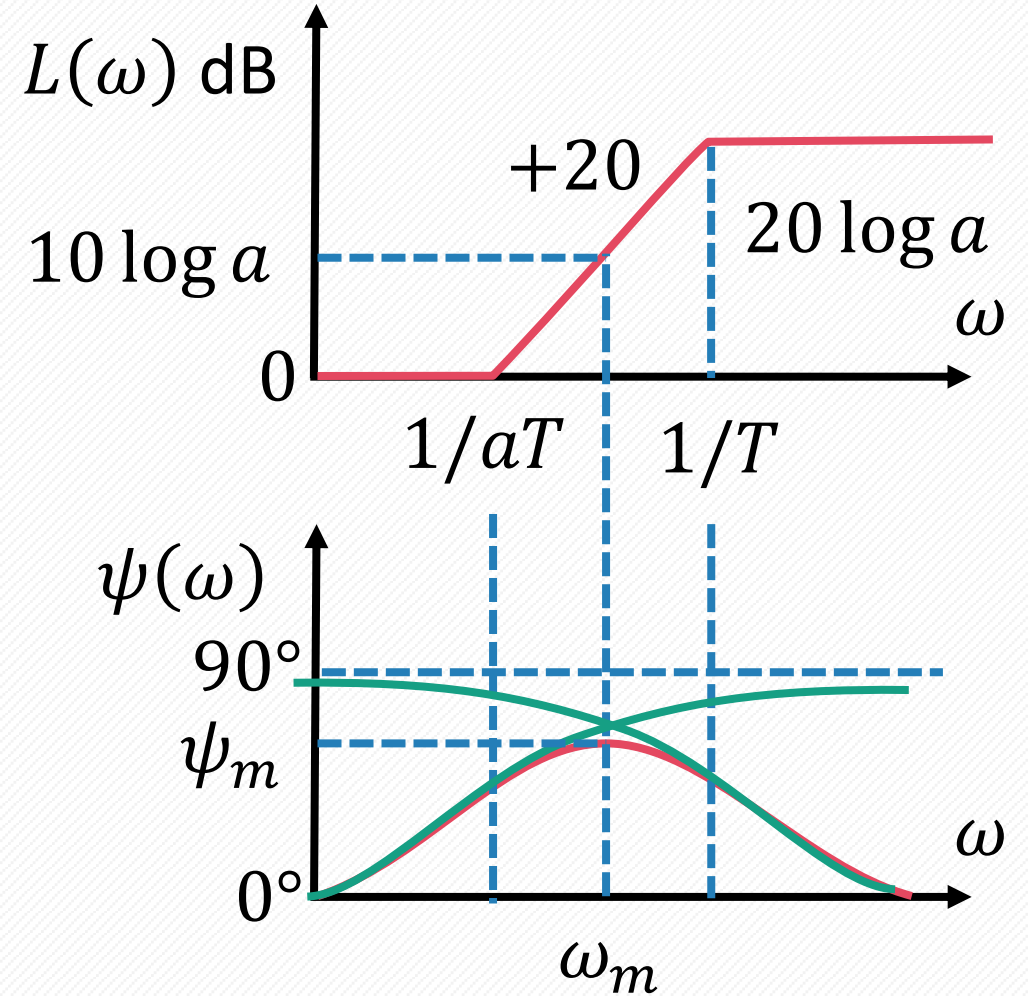
$$a \cdot G_s(s) = \frac{aTs + 1}{Ts + 1} \quad (a > 1)$$

抬升 $H = \lim_{\omega \rightarrow \infty} 20 \log |G(j\omega)| = 20 \log a$

$$\begin{aligned} \psi(\omega) &= \arctan(aT\omega) - \arctan(T\omega) \\ &= \arctan \frac{T\omega(a-1)}{1 + aT^2\omega^2} \end{aligned}$$

$$\frac{d\psi(\omega)}{d\omega} = 0 \rightarrow \frac{d}{d\omega} [\tan \psi(\omega)] = 0$$

$$\rightarrow \frac{d}{d\omega} \left[\frac{T\omega(a-1)}{1 + aT^2\omega^2} \right] = \frac{T(a-1)(1 - aT^2\omega^2)}{(1 + aT^2\omega^2)^2} = 0 \rightarrow \omega_m = \frac{1}{\sqrt{a}T}$$



超前网络特性

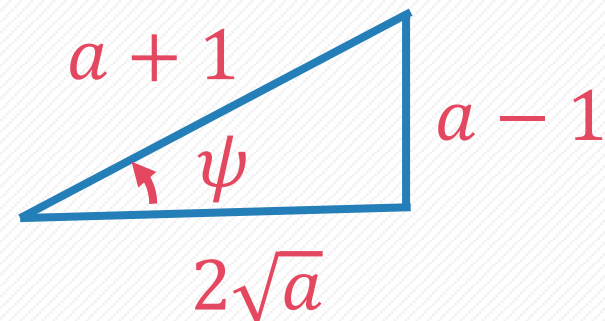
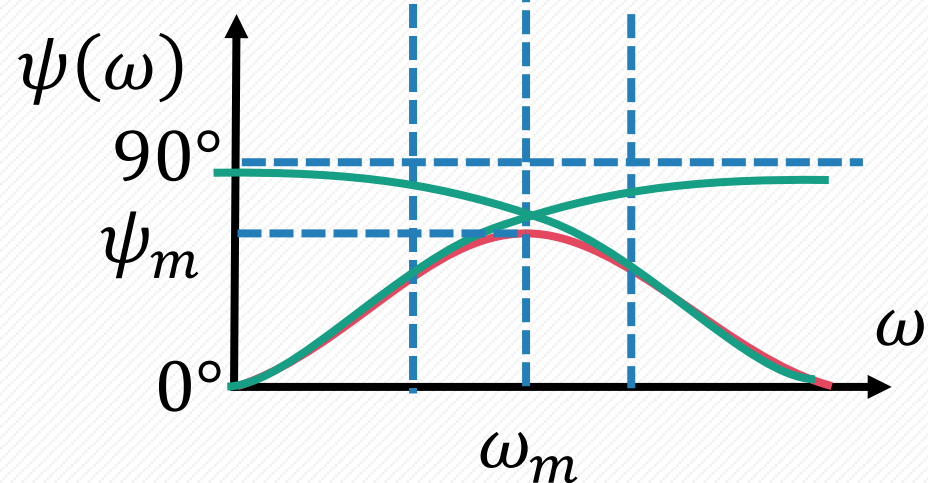
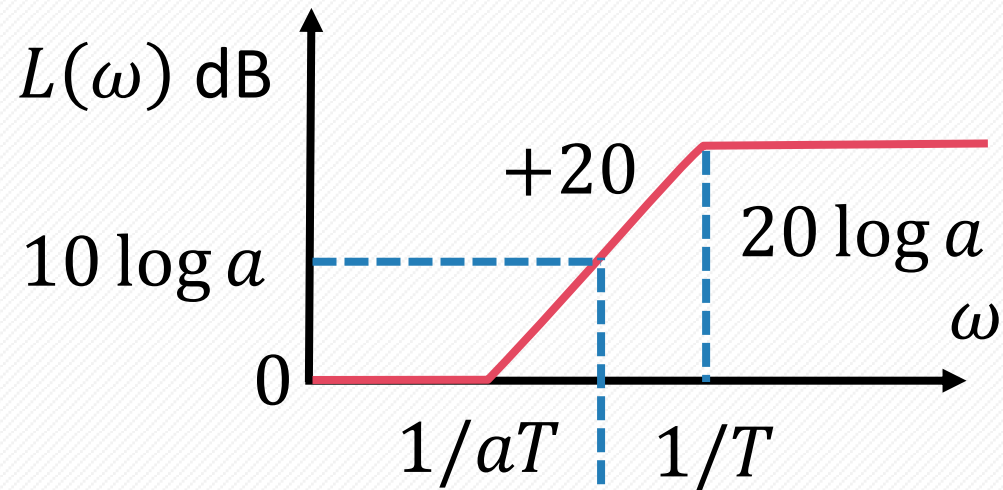
$$\omega_m = \frac{1}{\sqrt{a}T}$$

$$\tan \psi(\omega_m) = \left. \frac{T\omega(a-1)}{1+aT^2\omega^2} \right|_{\omega_m = \frac{1}{\sqrt{a}T}} = \frac{a-1}{2\sqrt{a}}$$

$$\psi_m = \psi(\omega_m) = \arctan \frac{a-1}{2\sqrt{a}} = \arcsin \frac{a-1}{a+1}$$

→

$$\begin{cases} a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m} \\ L(\omega_m) = 10 \log a \end{cases}$$



超前网络特性

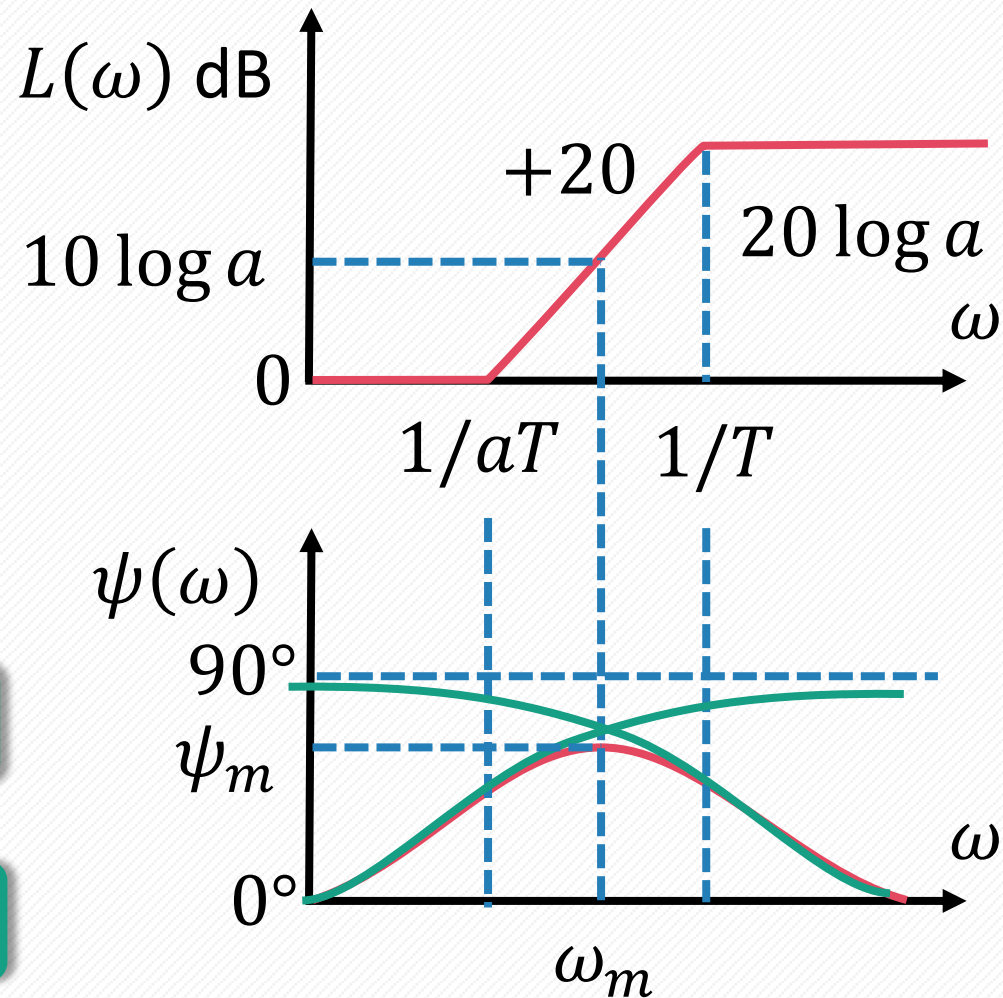
$$a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m}$$

$$L(\omega_m) = 10 \log a$$

1 超前网络特点:相角超前, 幅值增加

2 一级超前网络最大超前角为 60°

3 最有效的 $a \in (4, 10)$



步骤

实质：利用超前网络相角超前特性提高系统的相角裕度

给定 e_{ss}^* , ω_c^* , γ^*

1 由 $e_{ss}^* \rightarrow K$

2 由 $G_0(s) \rightarrow L_0(\omega) \rightarrow \omega_{c0} \rightarrow \gamma_0$: ω_{c0} , γ_0 均不足

3 确定 $\psi_m = \gamma^* - \gamma_0 + (5^\circ \sim 10^\circ)$

$$a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m} \quad L(\omega_m) = 10 \log a$$

4 作图确定原系统幅值为 $-10 \log a$ 时的频率

$$L(\omega_c) = -10 \log a$$

$$\omega_c = \omega_m \rightarrow \omega_2 = \omega_m \sqrt{a} = a \omega_1 \rightarrow G_c(s)$$

5 $G(s) = G_c(s)G_0(s)$ 验算 ω_c , γ 是否满足要求

例1 超前网络设计

$$G_0(s) = \frac{K}{s(0.2s + 1)}$$

要求: 1) $e_{ssv} \leq 0.32\%$, 2) $\gamma \geq 45^\circ$

1 由 $e_{ss}^* \rightarrow K$ $K = K_v = 316 \text{ s}^{-1}$

2 由 $G_0(s) \rightarrow L_0(\omega) \rightarrow \omega_{c0} = 39.6 \frac{\text{rad}}{\text{s}} \rightarrow \gamma_0 = 7.2^\circ$

3 确定 $\psi_m = \gamma^* - \gamma_0 + (5^\circ \sim 10^\circ) = 45^\circ - 7^\circ + 5^\circ = 42.8^\circ$

4 估算参数 $a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m} = \frac{1 + \sin 42.8^\circ}{1 - \sin 42.8^\circ} = 5.24$

取设计 $\omega_c = \omega_m$, 此时超前网络幅值 $10 \log a = 7.2 \text{ dB}$

原系统和超前网络在 ω_c 时幅值和应恰好为0。因此 $L_0(\omega_m) = -7.2 \text{ dB}$

由图 $L_0(\omega)$ 求得 $\omega_m = \omega_c = 60 \text{ rad/s}$

例1 超前网络设计

$$G_0(s) = \frac{K}{s(0.2s + 1)}$$

要求: 1) $e_{ssv} \leq 0.32\%$, 2) $\gamma \geq 45^\circ$

5 得出控制器

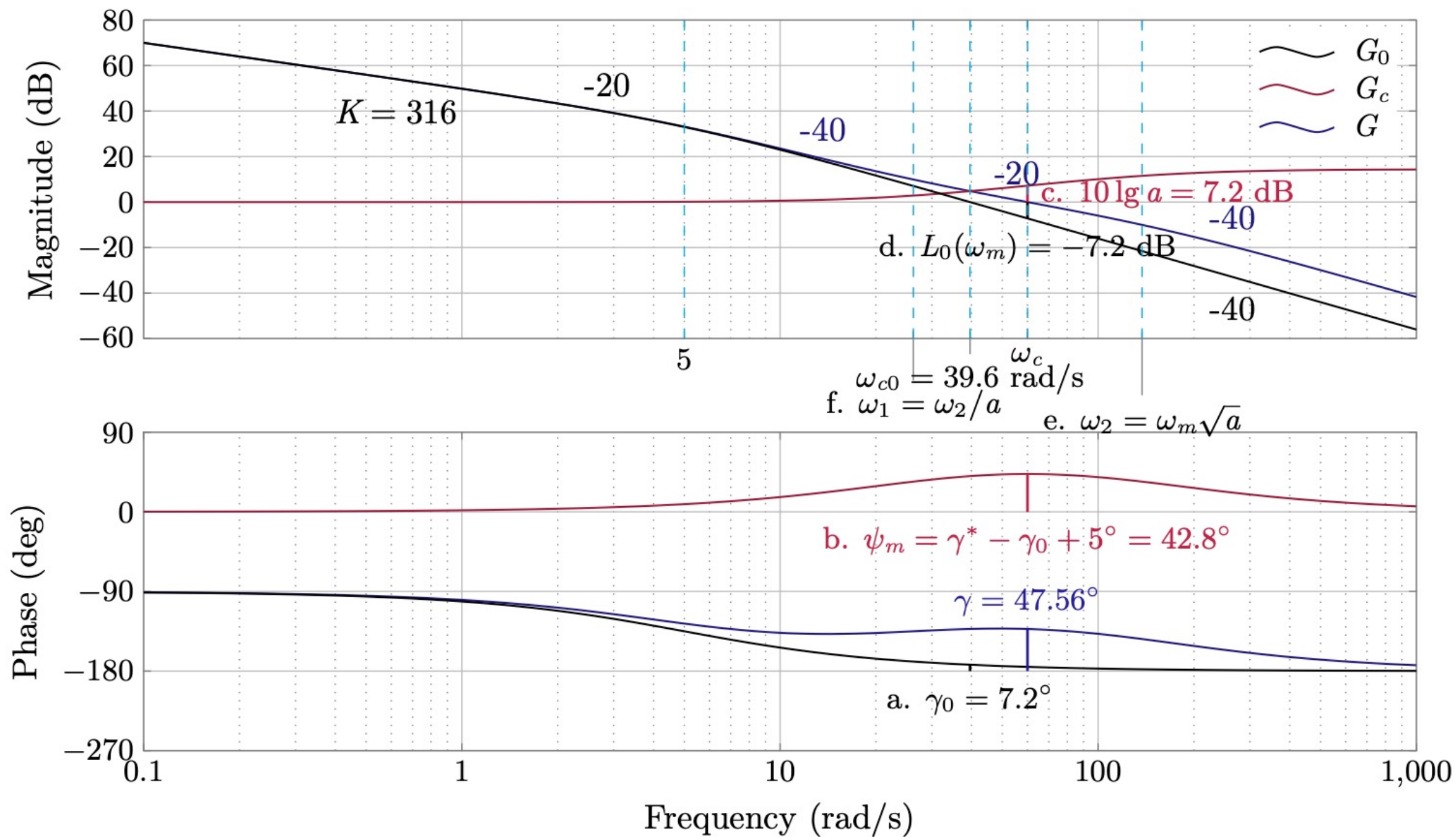
$$\omega_m = \frac{1}{\sqrt{a}T} \rightarrow T = 0.0073 \text{ s} \rightarrow \omega_2 = \frac{1}{T} = 137.42 \text{ rad/s}$$

$$\omega_1 = \frac{1}{aT} = 26.23 \text{ rad/s} \quad G_c(s) = \frac{0.0381s + 1}{0.0073s + 1}$$

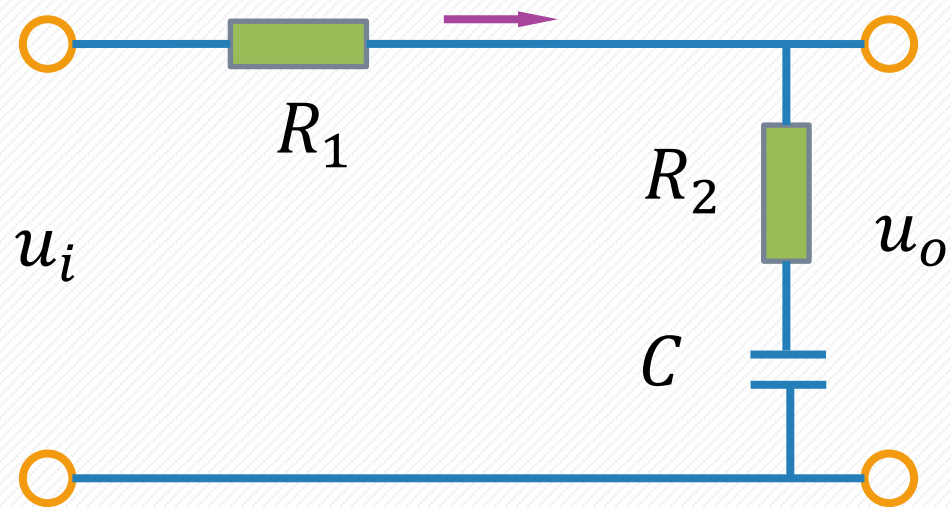
6 验算

$$G(s) = G_c(s)G_0(s)$$

$$\left\{ \begin{array}{l} \omega_c = 60 \text{ rad/s} \\ \gamma = 47.56^\circ \end{array} \right. \quad \text{满足要求}$$

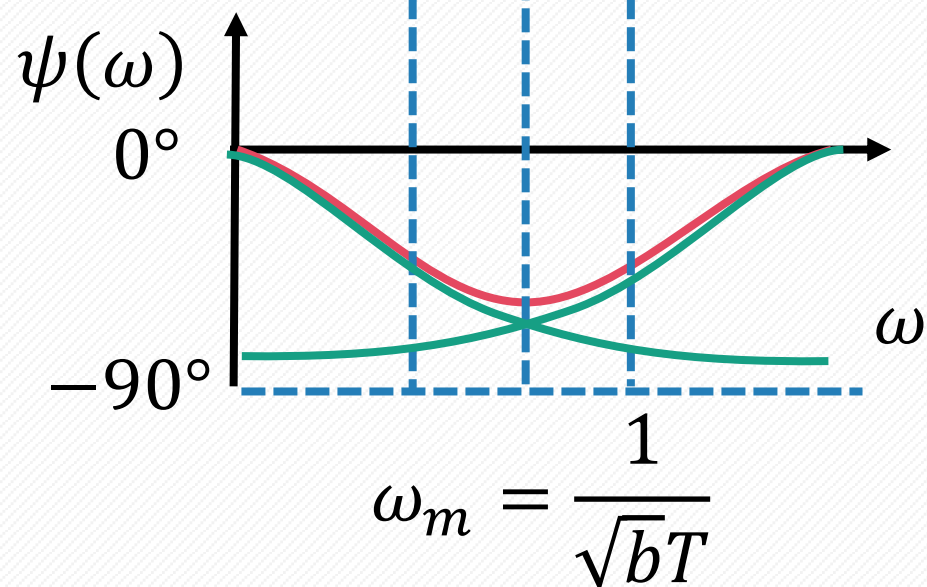
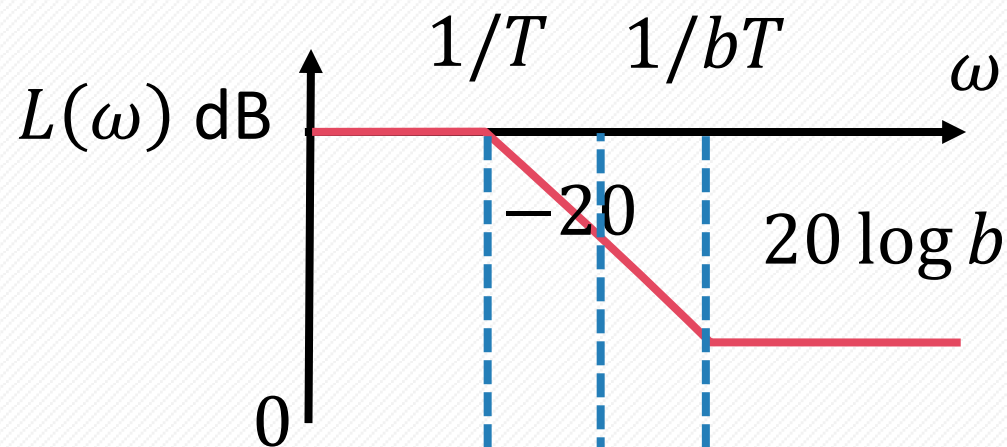


滞后网络特性



$$G_C(s) = \frac{U_o}{U_i} = \frac{R_2 + \frac{1}{Cs}}{R_1 + R_2 + \frac{1}{Cs}} = \frac{R_2 Cs + 1}{(R_1 + R_2)Cs + 1}$$
$$= \frac{bTs + 1}{Ts + 1}$$

$$T = (R_1 + R_2)C \quad b = \frac{R_2}{R_1 + R_2} < 1$$



滞后网络特性

1

幅频高频衰减特性，使原系统截止频率 ω_c 左移减小，相角裕度提高。适用于 ω_c 有余，相角裕度不足时；

2

相位滞后，会减小原系统相角裕度，应附加相角 $5 \sim 12^\circ$ ，并力求避免 ω_m 出现在 ω_c 附近(可放置在相角较为充裕的频率)，一般取

$$\omega_2 = \frac{1}{bT} = (0.1 \sim 0.2)\omega_c$$

相角滞后

是不利因素，应当避免

幅值衰减

是有利因素，应当利用

步骤

实质：利用滞后网络幅值衰减特性挖掘系统自身的相角储备

给定 e_{ss}^* , ω_c^* , γ^*

1 由 $e_{ss}^* \rightarrow K$

2 由 $G_0(s) \rightarrow L_0(\omega) \rightarrow \omega_{c0} \rightarrow \gamma_0$: ω_{c0} 有余, γ_0 不足

3 $\gamma_c(\omega) = \gamma^* + 5^\circ \leq 180^\circ + \psi(\omega_c) \rightarrow \omega_c$

4 参数 b 由 ω_c 时原系统需要降低的幅值决定

$$-L_0(\omega_c) = 20 \log b$$

5 作图设计 $\omega_2 = 0.1\omega_c$, $\omega_1 = b\omega_2$

6 $G(s) = G_c(s)G_0(s)$ 验算 ω_c , γ 是否满足要求

例2 滞后网络设计

$$G_0(s) = \frac{K}{s(0.2s + 1)}$$

要求: 1) $e_{ssv} \leq 0.32\%$, 2) $\gamma^* \geq 45^\circ$

1 由 $e_{ss}^* \rightarrow K$ $K = K_v = 316 \text{ s}^{-1}$

2 由 $G_0(s) \rightarrow L_0(\omega_c) \rightarrow \omega_{c0} = 39.6 \text{ rad/s} \rightarrow \gamma_0 = 7.2^\circ$ 不足

3 在 $L_0(\omega)$ 上找 $\gamma_0(\omega_c) = \gamma^* + 5^\circ = 45^\circ + 5^\circ = 50^\circ$

$$\gamma_0 = 180^\circ - 90^\circ - \arctan 0.2\omega \geq 50^\circ \rightarrow \omega_c = 4.05 \text{ rad/s}$$

4 确定参数 b 由图 $L_0(\omega_c = 4.05) = 38 \text{ dB}$, 令滞后网络

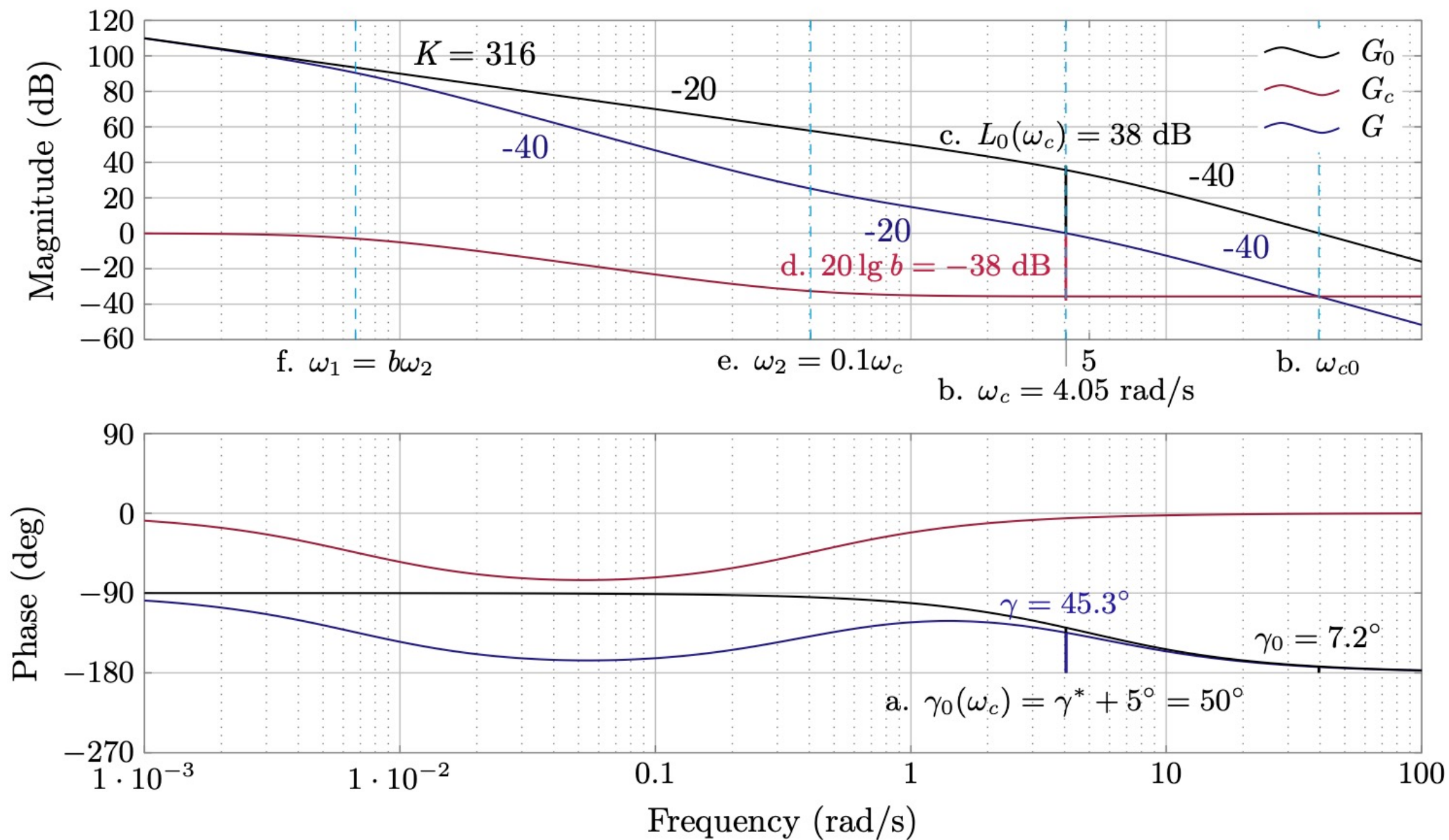
$$20 \log b = -38 \text{ dB} \rightarrow b = 0.0165,$$

$$\omega_2 = 0.1\omega_c = 0.4 \text{ rad/s}, \omega_1 = b\omega_2 = 0.0067 \text{ rad/s}$$

$$G_c(s) = \frac{2.5s + 1}{149.25s + 1}$$

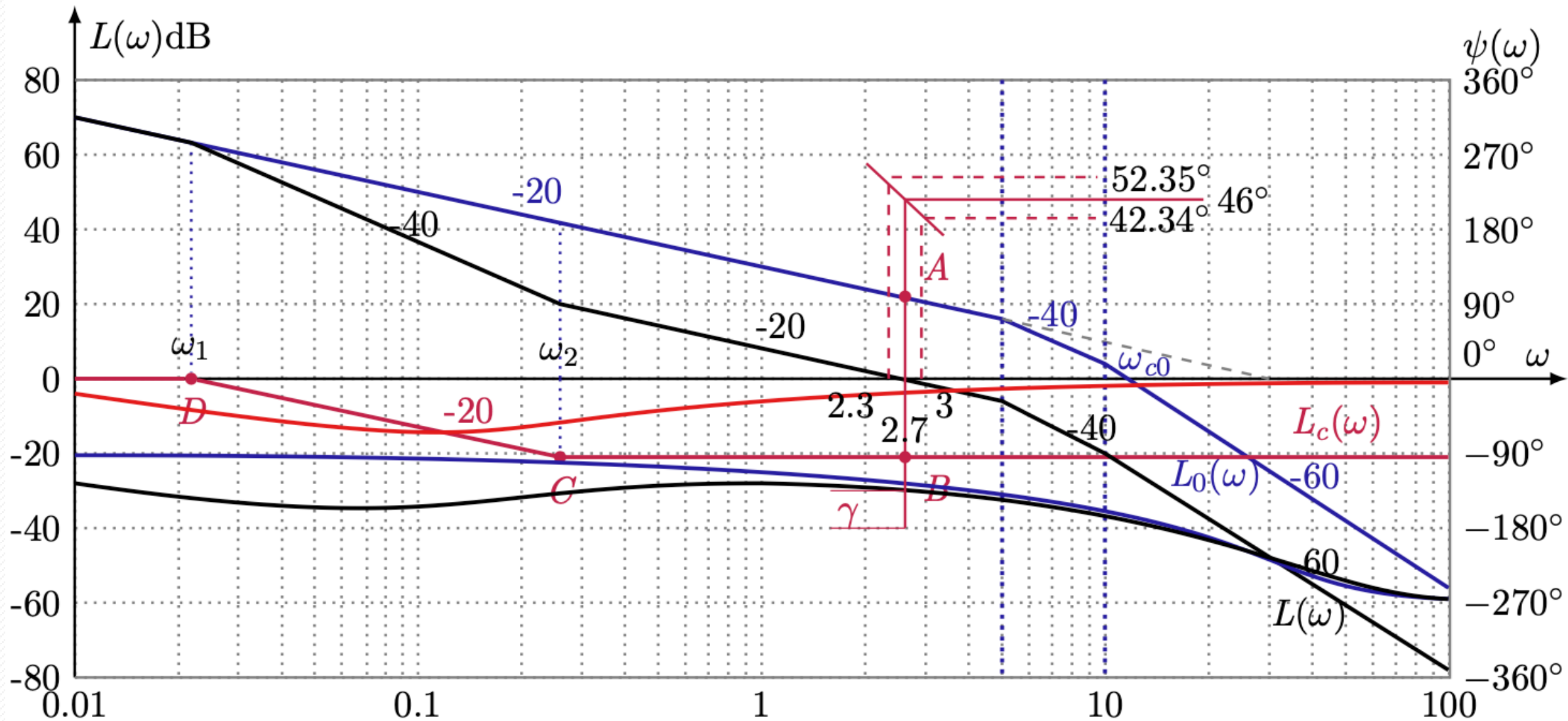
5 验算

$$\omega_c = 4.06 \text{ rad/s} \quad \gamma = 45.3^\circ \quad \text{满足要求}$$



例3 滞后网络设计

$$G_0(s) = \frac{K}{s\left(\frac{s}{5}+1\right)\left(\frac{s}{10}+1\right)}$$



例4 滞后网络设计

$$G_0(s) = \frac{K}{s(s+1)(s+2)}$$

要求: 1) $e_{ssv} \leq 0.2$, 2) $\gamma^* \geq 40^\circ$, 3) $h^* \geq 10$ dB

